

The Impact of Rating Visibility on Hotel Evaluations Across Service Scenarios

1. Introduction

Online rating systems have become a fundamental source of information for consumers' decision-making and brand reputation in the hospitality industry. Although previous research has examined how online reviews affect business outcomes (Xie et al., 2014) or acknowledges that transparency can induce social desirability bias (Jinyang Zheng, Guopeng Yin, Yong Tan, Jianing Ding, 2023), the interplay between rating visibility and specific service contexts is not fully discovered. Specifically, whether social pressure significantly influences rating behavior across positive, neutral, and negative scenarios remains an open question.

This study examines whether the disclosure of user ratings (public versus private) affects hotel evaluations across positive, neutral, and negative service scenarios. Using linear mixed-effects models and bootstrap resampling, we aim to determine whether social visibility systematically alters rating distributions or whether service quality remains the primary driver of consumers' evaluations.

2. Research Design

2.1 Experimental Setup

To identify the causal effect of rating visibility on consumer evaluation, this study employs a 2 (Visibility: Public vs. Private) $\times$ 3 (Scenario: Positive, Neutral, Negative) mixed factorial design. The factor Visibility serves as the between-subjects variable, while Scenario acts as the within-subjects variable.

2.2 Procedure

Participants were randomly assigned to either the treatment group (public rating) or the control group (private rating). In the treatment group, participants were informed that their ratings would be publicly displayed and linked to their personal identities, whereas the control group was assured of complete anonymity. Subsequently, each participant evaluated three different hotel service scenarios designed to elicit positive, neutral, and negative satisfaction levels.

3. Data

3.1 Data Collection

The data for this study was collected using an online survey hosted on the Qualtrics platform and mainly distributed via the online forum Dcard. After data cleaning to ensure response quality, the final sample consists of 77 participants, each contributing three observations, resulting in a total of 231 observations.

3.2 Descriptive Statistics

Table 1 presents the demographic characteristics of the participants. The sample is predominantly female (70%). Regarding age, the distribution heavily favors young adults, with over 90% of participants aged between 18 and 34. Consequently, the findings of this study primarily reflect the perspectives of a younger population.

Table 1. Demographic Characteristics of Participants	
Characteristic	N = 77
Gender	
Female	54 (70%)
Male	21 (27%)
Prefer not to say	2 (2.6%)
Age Group	
18-24	25 (32%)
25-34	46 (60%)
35-44	3 (3.9%)
45-54	1 (1.3%)
Above 55	1 (1.3%)
Under 17	1 (1.3%)

4. Methodology

4.1 Model Specification

To examine the causal impact of rating visibility on user scores across different service contexts, and to address the repeated-measures structure of the data in which observations from the same participant are not independent, we utilize a linear mixed-effects model that captures individual-specific variation in rating behavior. The model is specified as follows:

$$Score_{ij} = \beta_0 + \beta_1 Treatment_i + \beta_2 Scenario_j + \beta_3 (Treatment_i \times Scenario_j) + u_i + \epsilon_{ij}$$

The variables in this model are defined as follows:

- $Score_{ij}$ represents the rating given by participant i for scenario j .
- $Treatment_i$ is a binary indicator variable (1 = Public/Treatment, 0 = Private/Control).
- $Scenario_j$ represents the categorical service levels (Reference category = Neutral).
- $Treatment_i \times Scenario_j$ represents the interaction effect, testing whether the impact of visibility varies across service quality level.
- u_i represents the random intercept for participant i .
- ϵ_{ij} represents the residual error term.

4.2 Estimation and Inference

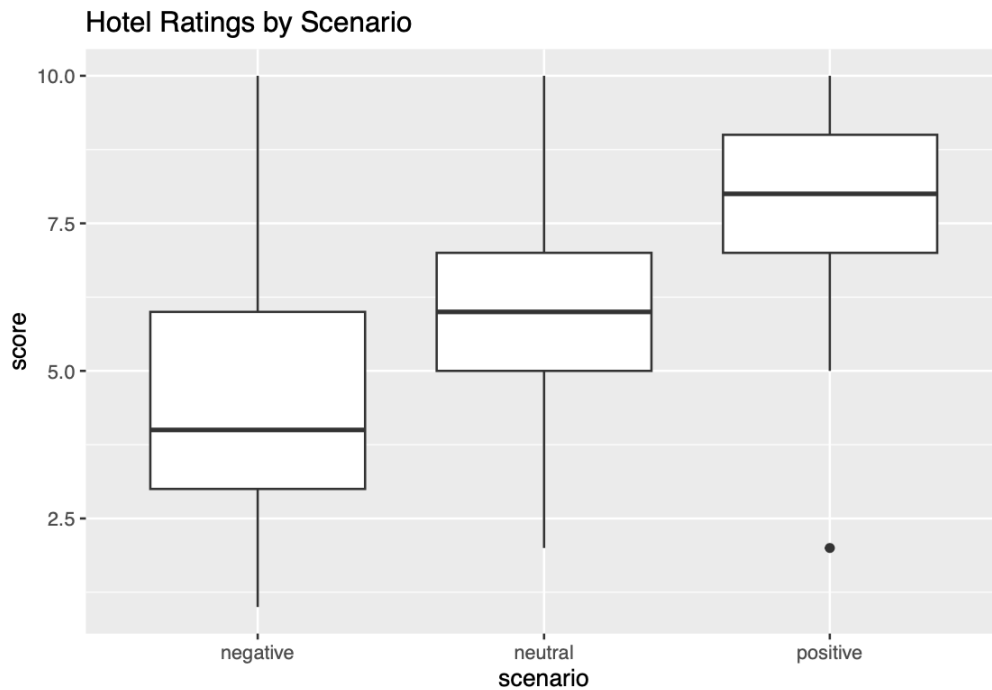
The model is estimated using Restricted Maximum Likelihood (REML) through the lme4 package in R, which provides unbiased estimates of variance components. Model fit is evaluated using Marginal R^2 , which reflects the variance explained by fixed effects. Conditional R^2 reflects the variance explained by both fixed and random effects.

5. Results

5.1 Exploratory Data Analysis

After collecting and processing the data into a long format, we began our exploratory data analysis (EDA) by visualizing the score distributions across the three scenarios. First, we employed boxplot to examine whether the participants' ratings aligned with the intended experimental design. This serves as a manipulation check to ensure the scenarios were perceived as expected.

Figure 1: Boxplot of scores across scenarios



As illustrated in the boxplot, the scores for the positive, neutral, and negative scenarios follow a clear descending order. This distinct distribution suggests that the experimental manipulation of the hotel scenarios was effective, as participants evaluated the scenes aligning perfectly with our design expectations.

To further investigate the potential interaction between the treatment and the scenarios, we categorized the boxplots by group (Experimental vs. Control). This allowed for an initial exploration of whether the rating publicly had a varying impact across different contexts.

Figure 2: Grouped boxplot by treatment



As illustrated in the boxplot, the score distributions for the treatment and control groups appear visually similar in the positive and negative scenarios. However, a noticeable gap was shown in the neutral scenario, where the treatment group (public exposure) provided lower average ratings than the control group. This visual pattern suggests that participants might adopt a stricter evaluation standard when they are aware that their reviews will be publicly accessible, particularly when evaluating hotels in a neutral (middle) scenario.

5.2 Linear Mixed-effect Model

After confirming the successful manipulation, we continued to evaluate the primary interaction effect between treatment and scenario. As explained in the Methodology section, we fit the data using a linear mixed-effects model, with participants' baseline evaluation standards specified as random effects. The results are presented below.

Table 1: Fixed effects of the LMM

effect	term	estimate	std.error	statistic	df	p.value
fixed	(Intercept)	6.615	0.266	24.894	174.467	0.000
fixed	treatment1	-0.563	0.378	-1.488	174.467	0.139
fixed	scenariopositive	1.205	0.296	4.074	150.000	0.000
fixed	scenarionegative	-2.462	0.296	-8.322	150.000	0.000
fixed	treatment1:scenariopositive	0.505	0.421	1.200	150.000	0.232
fixed	treatment1:scenarionegative	0.646	0.421	1.534	150.000	0.127

Since the neutral scenario serves as our baseline, the model's intercept is estimated at 6.615 ($p < 0.05$). This value represents the average rating participants assigned to the neutral hotel scene when reviewing privately and anonymously.

Regarding scenario manipulations, the positive scenario received, on average, 1.2 units higher than the neutral baseline (scenariopositive), while the negative scenario received 2.5 units lower than the neutral baseline (scenarionegative). Both effects are statistically significant ($p < 0.05$), providing robust evidence that our experimental manipulation of hotel scenarios was highly effective.

In terms of the treatment effect (treatment1), the estimate of -0.563 represents the average difference in rating scores within the neutral scenario between the experimental and control groups. However, this difference is not statistically significant ($p = 0.139$). Therefore, the model suggests the nonexistence of a main treatment effect, indicating that the disclosure of ratings to the public does not substantively influence participants' evaluative behavior.

Regarding the interaction effects, the primary focus of this study, the estimate for the positive scenario (treatment1:scenariopositive) is 0.505. This value suggests that in the positive hotel context, the treatment effect (public vs. private) is 0.505 units higher than it is in the neutral context. However, this interaction was not statistically significant ($p = 0.232$). Similarly, for the negative hotel scene, the interaction effect was also non-significant. These results indicate that the effect of disclosing the rating purpose does not exist as a main effect, nor does it vary significantly across different contexts.

Table 2: R^2 of the LMM

	x		x
Conditional R2	0.657	Marginal R2	0.446

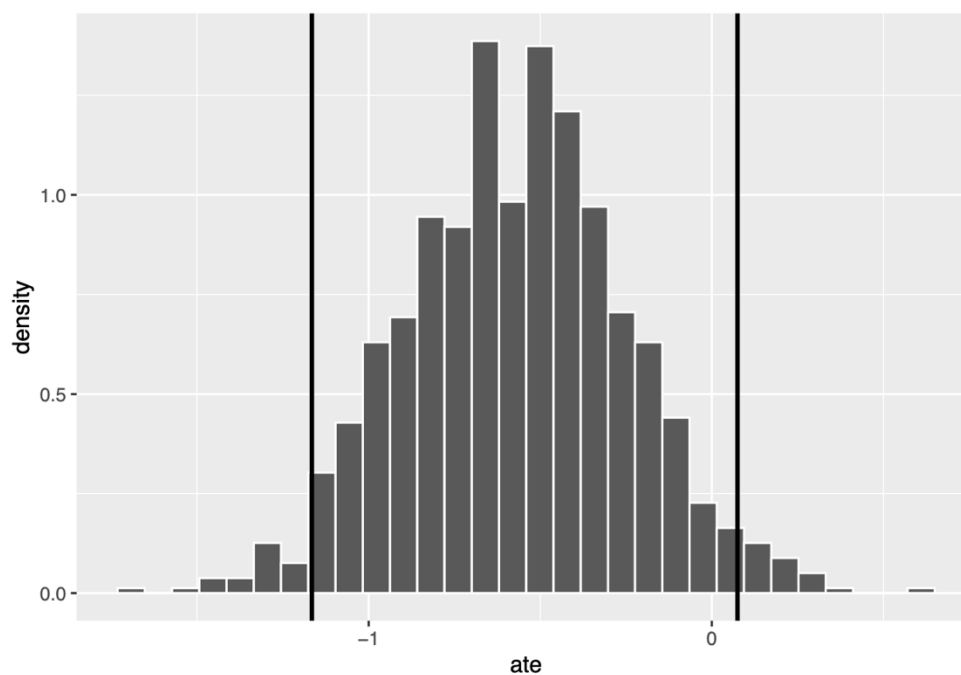
We also examined the model fit using R^2 measures for mixed-effects models. The marginal R^2 was estimated to be 0.446, indicating that the fixed effects—treatment (public vs. private) and scenario—together explain approximately 44.6% of the variance in hotel ratings. Given that the main treatment effect was not statistically significant, this explained variance is largely attributable to differences across scenarios.

The conditional R^2 was estimated to be 0.657, suggesting that when participant-level random effects are taken into account, the model explains about 65.7% of the total variance in the data. This increase reflects the contribution of individual differences in baseline rating standards across participants.

5.3 Bootstrap for Descriptive Robustness Check in the Neutral Hotel Scenario

The linear mixed-effects model indicates that the treatment effect does not exist across the three hotel scenarios, there is theoretically no need to test the treatment effect within each scenario separately. However, during the EDA stage, a more noticeable difference in ratings between the treatment and control groups was observed in the neutral hotel scenario. Therefore, we conducted a simple descriptive robustness check using bootstrap resampling for this scenario.

Figure 3: Bootstrap results



According to the bootstrap distribution of the ATE, the 95% confidence interval was estimated to be $(-1.21, 0.071)$. Since this interval includes zero, the results indicate no observable treatment effect in the neutral hotel scenario. In conclusion, participants do not appear to exhibit distinct rating behavior when evaluating a hotel in a more ambiguous context, even when they are aware that their reviews may be publicly disclosed.

6. Discussion & Conclusion

6.1 Results Discussion

Many prior studies have shown that individuals often behave differently in public settings than in private contexts. People tend to become more cautious when expressing their opinions publicly due to increased social visibility and accountability. Therefore, we expected to observe distinct patterns in hotel rating behavior when reviews were publicly disclosed.

However, our experimental results suggest otherwise. Participants did not exhibit significant differences in hotel ratings between public and private review conditions. Moreover, this consistency in ratings did not vary across relatively positive, neutral, or negative hotel scenarios.

6.2 Experiment Limitation

Despite the non-significant findings, this study has several major limitations that should be acknowledged.

First, the lack of participant diversity should be acknowledged. Due to time and resource constraints, participants were primarily recruited from a Taiwanese online college student forum, Dcard. Although the platform is popular, its user base limits the demographic diversity of our sample: approximately 70% of participants were female, and nearly 60% were between the ages of 25 and 34. This relatively homogeneous sample may have influenced our research results.

Second, although participants were provided with a review context description at the beginning of the experiment to indicate their assigned condition (public vs. private), it is possible that the treatment manipulation was not strong enough compared to the influence of the three hotel scenarios. Participants may have focused more on the scenario itself rather than the public or private nature of the review.

Lastly, the description of the hotel scenarios may not have been detailed enough. Each scenario was designed to convey the overall experience and feeling of the hotel, but did not account for the fact that different service aspects may carry different weight in participants' evaluations. For example, some participants might consider hotel breakfast unimportant when rating a hotel, while others might view it as very

important. Variation in the emphasis of different service items across scenarios could therefore be a confounding factor in our study.

6.3 Conclusion and Future Improvement

Since the experiment was designed and conducted within a short time period, there remain several limitations and compromises that were not fully addressed. Overall, the study suggests that the public disclosure of reviews does not significantly affect participants' hotel rating behavior across positive, neutral, and negative scenarios.

For future improvements, expanding the sample to include a broader range of demographics would enhance the generalizability of the findings. Additionally, the experimental design could be made more detailed to better capture participants' authentic reactions. Further research could also explore the psychological mechanisms underlying review behavior to provide deeper insights.

Reference

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Appendix: Manipulation Description in the Experimental Questionnaire

Control group:

Please imagine that you are a consumer who has just completed a stay and are filling out a feedback form for the hotel service staff. Please provide an honest rating and feel free to express your true feelings and preferences. All ratings and responses will be completely anonymous and confidential and will be used by the hotel solely as a basis for service improvement.

Treatment group:

Please imagine that you are a consumer who has just completed a stay and is leaving a rating for the following accommodation experience. The score and review you provide will be publicly posted under your personal account on a major travel platform or search website, and your feedback will serve as an important reference for potential consumers when making their decisions.

Neutral scenario

You have just completed a two-night stay at a tourist hotel. The room was newly decorated, and the rooftop outdoor pool was excellent, providing a very pleasant experience. However, the breakfast quality was inconsistent over the two days: the first day was good, but the food was cold on the second day. The air conditioner in the room was occasionally noisy. The hotel is conveniently located, but the parking lot requires a fee. When you reported the air conditioning issue, the staff were friendly, although they were unable to resolve it immediately.

Positive scenario

You have just completed a two-night stay at a chain hotel. The room was extremely spacious and well-equipped, with a premium mattress. The staff were friendly and provided fast and attentive service, making you feel very valued. However, the minibar price labels in the room were incorrect, which caused some confusion at checkout (although the issue was eventually resolved). The hotel's rates were noticeably higher than the average in the area. The lobby was grand in design, and both check-in and check-out were completed within two minutes.

Negative scenario

You have just completed a two-night stay at a business hotel. The room was clean, and the bed was very comfortable, allowing you to rest well after a tiring workday.

However, the soundproofing was poor, and on the first night you were awakened twice by noise in the hallway. During check-out, the front desk staff were indifferent and unfriendly, and the process took about 10 minutes. The hotel provided a complimentary breakfast, which was simple but acceptable.